

# Sharanya Mishra

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## EDUCATION

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|--------------------------------------------------------------------------------------|---------------------------------|
| <b>Illinois Institute of Technology</b><br>Master of Science, <i>Data Science</i>    | Aug 2024 - May 2026<br>GPA: 3.7 |
| <b>PES University</b><br>Bachelor of Technology, <i>Computer Science/Engineering</i> | Dec 2020 - Jun 2024<br>GPA: 3.5 |

## SKILLS

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| <b>Programming Languages:</b> Python, SQL, R, JavaScript, C++                                                                                                                           |
| <b>Data Science &amp; Statistics:</b> Pandas, NumPy, scikit-learn, Statistical Analysis, Hypothesis Testing, Regression, Time-Series Analysis, Forecasting, Monte Carlo Simulation, NLP |
| <b>Analytics &amp; Visualization:</b> Tableau, Power BI, Excel, Matplotlib, Seaborn, Plotly                                                                                             |
| <b>Machine Learning:</b> PyTorch, TensorFlow/Keras, YOLO, ResNet, ARIMA, GARCH, Model Evaluation                                                                                        |
| <b>Databases &amp; Data:</b> PostgreSQL, Supabase, pgvector                                                                                                                             |
| <b>Cloud &amp; Tools:</b> AWS, Azure, Git, Docker, Jupyter                                                                                                                              |

## WORK EXPERIENCE

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| <b>Lead Research Assistant</b>   Illinois Institute of Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Jan 2025 – Jun 2026 |
| <ul style="list-style-type: none"><li>Analyzed solver behavior across <b>50+ optimization configurations</b> using Python, comparing convergence, runtime and computational efficiency.</li><li>Built automated <b>Python</b> workflows to run experiments and streamline benchmarking, reducing simulation runtime by ~30%.</li><li>Developed end-to-end research tooling that transformed optimization outputs into interactive visualizations and interpretable results for technical decision-making.</li></ul>                              |                     |
| <b>Data Science Consultant / Project Lead</b>   LabelMaster                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Jan 2026 - May 2026 |
| <ul style="list-style-type: none"><li>Built <b>SQL and Python</b> ETL pipelines to clean, transform and validate <b>14 years of customer</b>, transaction, product and location data.</li><li>Built <b>customer segmentation</b> and cohort analyses across <b>4 integrated datasets</b>, surfacing high-value customers and win-back opportunities.</li><li>Used <b>Pandas and Matplotlib</b> to visualize customer segments, behavioral patterns and revenue trends, translating findings into retention and growth recommendations.</li></ul> |                     |
| <b>Data Classification Intern</b>   Aptos Retail                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jan 2024 - Jul 2024 |
| <ul style="list-style-type: none"><li>Built <b>15+ Azure-based data validation workflows</b> using Microsoft Purview and Salesforce to improve reporting accuracy and data consistency.</li><li>Automated data-quality checks across <b>5+ teams</b>, reducing manual validation effort and improving consistency of downstream analytics.</li><li>Investigated and reconciled cross-system data issues, standardizing classification logic to improve reliability for reporting and analysis.</li></ul>                                         |                     |
| <b>Data Analyst Intern (Risk Advisory)</b>   Deloitte Touche Tohmatsu LLC                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Jun 2023 - Aug 2023 |
| <ul style="list-style-type: none"><li>Built <b>Python and SQL</b> workflows to clean, validate and analyze <b>7+ client datasets</b>, detecting anomalies and reporting inconsistencies.</li><li>Performed variance and trend analysis and built Excel dashboards using Pivot Tables and VLOOKUP to support client risk decisions.</li><li>Investigated data-quality issues across multi-source datasets to support accurate reporting and audit analysis.</li></ul>                                                                             |                     |
| <b>Data Analyst Intern</b>   Capgemini                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Jun 2022 - Aug 2022 |
| <ul style="list-style-type: none"><li>Analyzed client data using SQL and Tableau to identify demand trends, performance patterns and business drivers.</li><li>Built <b>Tableau dashboards</b> on transformed datasets to give stakeholders a reusable view of demand and performance trends.</li></ul>                                                                                                                                                                                                                                          |                     |

## PROJECTS

### HerWay - AI-Powered Urban Analytics Platform

- Integrated and analyzed **150K+ 311 records, crime data and 1,000+ community posts** across **77 Chicago communities**.
- Analyzed neighborhood-level patterns across public-service complaints, crime and community sentiment to surface differences in local safety and resident experience.
- Applied **NLP and geospatial analytics** to combine structured and unstructured data into interactive dashboards and actionable neighborhood insights.

### ForeQuest - AI-Powered Financial Forecasting Platform

- Developed ARIMA and GARCH time-series models on **10+ years of market data** to forecast stock trends and volatility.
- Achieved **~85% R<sup>2</sup> and <0.02 MAE** in volatility prediction while integrating **Groq-hosted LLMs** to deliver explainable forecast interpretation, risk analysis, and conversational financial insights through a deployed Streamlit application.
- Applied Monte Carlo simulation and option-pricing models to quantify uncertainty, risk and potential financial outcomes.

### InsightAI - Production RAG Agent for Google Workspace

- Built a **RAG-based intelligence platform** using PostgreSQL, Supabase + pgvector, FastAPI and LangGraph to retrieve relevant information and generate grounded, confidence-aware responses.
- Built **multi-step agent workflows with LangGraph** to route queries and retrieve relevant context from connected data sources.
- Designed semantic indexing and retrieval workflows and deployed the end-to-end application on Vercel, Render and Supabase with OAuth 2.0 authentication and production database integration.

### Audio Description for the Visually Impaired | IEEE 2024

- Developed a multimodal deep-learning pipeline combining **object recognition, action recognition and text-to-speech** to generate real-time descriptions of video content.
- Evaluated generated descriptions quantitatively, achieving a **BLEU score of 0.70** while creating an assistive AI system for visually impaired users.
- Processed multimodal video data to support model training and evaluation.